EPILOGUE TO 40TH ANNIVERSARY EDITION

Scientists, unlike politicians, can take pleasure in being wrong. A politician who changes his mind is accused of 'flip-flopping'. Tony Blair boasted that he had 'not got a reverse gear'. Scientists on the whole prefer to see their ideas vindicated, but an occasional reversal gains respect, especially when graciously acknowledged. I have never heard of a scientist being maligned as a flip-flopper.

In some ways I would quite like to find ways to recant the central message of *The Selfish Gene*. So many exciting things are fast happening in the world of genomics, it would seem almost inevitable—even tantalizing—that a book with the word 'gene' in the title would, forty years on, need drastic revision if not outright discarding. This might indeed be so, were it not that 'gene' in this book is used in a special sense, tailored to evolution rather than embryology. My definition is that of George C. Williams, one of the acknowledged heroes of the book, now lost to us along with John Maynard Smith and Bill Hamilton: 'A gene is defined as any portion of chromosomal material that potentially lasts for enough generations to serve as a unit of natural selection.' I pushed it to a somewhat facetious conclusion: 'To be strict, this book should be called … *The slightly selfish big bit of chromosome and the even more selfish little bit of chromosome.'* As opposed to the embryologist's concern with how genes affect phenotypes, we have here the neo-Darwinist's concern with changes in frequencies of entities in populations. Those entities are genes in the Williams sense (Williams later called that sense the 'codex'). Genes can be counted and their frequency is the measure of their success. One of the central messages of this book is that the individual organism doesn't have this property. An organism has a frequency of one, and therefore cannot 'serve as a unit of natural selection'. Not in the same sense of *replicator* anyway. If the